Name: [redacted]

Enrolment No: [redacted]

UPES

UPES

End Semester Examination, December 2025

Course: Elements of AIML

Program: B.Tech (CSE)+BCA+BSc (CS)

Course Code: CSAI2018

Nos. of page(s) : 02

Semester: III

Time : 03 hrs.

Max. Marks: 100

Instructions: Kindly attempt according to the provided time and given weightage. Calculators are allowed.

SECTION A

(5Qx4M=20Marks)

S.No.	Please ensure that you answer all questions.Explain these section questions within 40-50 words at max.	Marks	CO
Q 1	What are AI techniques? Explain any two commonly used AI techniques.	4	CO1
Q 2	What are quantifiers? Give examples. What is the difference between ∀ and ∃?	4	CO2
Q 3	What are categorical, numerical, and ordinal data? Give examples.	4	CO3
Q 4	What is clustering? How does it differ from classification?	4	CO3
Q 5	Discuss the use of ML in early disease diagnosis with examples.	4	CO4

SECTION B

(4Qx10M= 40 Marks)

	Note that there is an OR question at Question 9 (attempt 9 (a) + 9 (b) OR only 9(c). Check minutely. Explain these section questions within 80-100 words at max.		CO
Q 6	Write the following statements in predicate logic and explain while writing in a brief way (not more than 2 lines):(5Q*2M=10 marks) **a)** "All fathers are parents." **b)** "Someone admires every teacher." **c)** "Every employee has a manager." **d)** "There exists a person who owns a car." **e)** "Some student knows every answer."	10	CO2
Q 7	Cluster the following 6 data points into 2 clusters using the K-Means algorithm. Initial Centroids: $C_1 = (3, 9)$ and $C_2 = (6, 3)$ Data Points: A(3,9), B(4,8), C(6,4) ,D(7,2) ,E(5,7), F(8,3) Perform one complete iteration of the K-Means algorithm. • Compute the Euclidean distance of each point from both centroids. • Assign each point to the nearest cluster. • Present the cluster assignment at the end of iteration.	10	CO3
Q 8	Describe the impact of AI in the banking industry and how it is used to enhance security and fraud detection.	10	CO3
Q 9	a)State the difference between PCA-ICA and explain each concept briefly. (6 marks)	10	CO4

	b) State the difference between Supervised Learning and Un-supervised Learning. **(4 marks)** **OR** **c)** Describe the concept of cross-validation and its importance in ensuring the reliability of Machine Learning models. **(10 marks)**		
SECTION-C **(2Qx20M=40 Marks)** **Note that there is an OR question at Question 10 (a and b) or 10 (c and d) and at Question 11 attempt any four questions. Check minutely. Explain these section questions within 100-120 words at max.**			
Q 10	**a)** Explain Artificial Intelligence and its major real-world applications. Discuss how AI is transforming sectors such as healthcare, education, industry, and transportation. **(10 marks)** **b)** What is a dataset in Machine Learning? Explain data handling steps such as data cleaning, feature selection, feature engineering, handling missing values, normalization, and encoding categorical data. **(10marks)** **OR** **c)**Explain the syntax and semantics of propositional logic. Discuss well-formed formulas, logical connectives, and truth-table evaluation with suitable examples. **(10 marks)** **d)**What are the common performance metrics for classification tasks in machine learning? Explain precision, recall, and F1-score. **(10 marks)**	**20**	**CO3, CO5**
Q 11	Write Short note : **(Any four)** **(4Q*5M=20 marks)** **a)** Define AI and discuss any two of AI Techniques. **b)** Linear discriminant Analysis (LDA). **c)** Explain the concept of feature sets. **d)** Explain the concept of dimensionality reduction. **e)** Discuss the concepts of bias and variance in supervised learning. **f)** AUC Curve.	**20**	**CO3, CO5**